

Essentials of Mixed and Longitudinal modeling - 2/2023

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Case study: Progression of quality of life in rectal cancer patients

1 Instructions

The aim of this assignment is for you to illustrate that: (1) you have sufficiently comprehended the basics of the course, (2) you are able to analyse on your own a dataset and (3) you are able to correctly interpret the analysis results.

You are expected to prepare a report answering the questions below. The case study report should not be read as annotated R output. It should comprise an introduction of the problem, a description of the data, the models and their assumptions, a description of inference, e.g., statistical tests used, results including relevant estimated effects with standard errors or confidence intervals. You may use an appendix to add a discussion on e.g, how the model assumptions have been checked, and the R code. Please add tables and graphs, where necessary, but show some restraint.

On the front of the report there should be a declaration that all students in the group have contributed sufficiently to the study and that the work (analysis, R programs and text) is original and not copied from elsewhere, signed by all students.

2 Introduction

The simulated data set considered in this assignment concerns a longitudinal study, in which the goal is to investigate the potential benefit of adjuvant chemotherapy after neoadjuvant (chemo-radiotherapy) and curative resection for rectal cancer. The potential benefit is evaluated in terms of improving the quality of life of patients measured via the EORTC-C30 questionnaires and the presence of complaints in at most two of the items of the symptoms/side effects scale.

In this study 226 patients with histologically proven stage II or III rectal cancer were assigned to adjuvant chemotherapy or observation after preoperative (chemo)radiotherapy and surgery. All patients were invited to fill in quality of life questionnaires at 1, 3, 6 and 12 months post-surgery.

A complication in this study is that not all planned measurements have been recorded for several reasons e.g., loss to follow-up, relocation, side-effects, etc.

3 Data file

- QOL.csv
- Variables
 - ID: patient indicator.
 - Time: Visit number i.e., 1-4.
 - Score: Quality of Life score at each of the 4 visits.
 - Treat: Treatment allocation (0: observation, 1: adjuvant chemotherapy).
 - Age: Age at inclusion in the study.
 - Complaints: Presence of complaints in more than 2 items (0: no, 1: yes).

4 Questions: Quality of Life score

1. Describe the data using summary statistics and graphical techniques to explore the mean structure, the variance structure and the correlation structure. Summarize your conclusions e.g., do you see non-linearities, differences between the groups, etc.
2. Fit a multivariate model. The main research questions are:
 - (a) Does the assigned treatment have an effect on the mean progression of the quality of life?
 - (b) Is there a difference in the quality of life at six months after surgery after completion of adjuvant chemotherapy for patients in the treatment group? Give the corresponding 95% confidence interval and p -value.

- (c) Is there a difference between the two groups at 12 months after surgery? Give the corresponding 95% confidence interval and p -value.

Which covariance structures are applicable in this case? What is the most parsimonious structure you can find? Find the most parsimonious mean structure which can be used to describe the average evolutions in the data. Run model diagnostics to check the validity of the model assumptions.

3. Formulate a plausible linear mixed-effects model and reply to the same questions as in the multivariate model above. What is the most parsimonious random-effects and mean structure that fits best on the data? Run model diagnostics to check the validity of the model assumptions.
4. Compare the results between the multivariate model and the linear mixed effects model. Which would you choose as your final model?

Present your analysis results in a manner understandable by a non-statistician; you may use graphical tools to communicate your findings.

5 Questions: Presence of Complaints

1. Describe the data using summary statistics and graphical techniques to explore the longitudinal evolutions. Summarize your conclusions.
2. Use the GEE approach. The main research question is:
 - Does the probability of at most 2 complaints change in time and how does this depend on the assigned treatment?
3. Answer the same questions but now using the mixed-effects logistic regression.
4. Compare the results between the GEE and the mixed-effects logistic regression model. Which would you choose as your final model?

Present your analysis results in a manner understandable by a non-statistician; you may use graphical tools to communicate your findings.

6 General remarks

- For each question, motivate your choice of techniques, estimation methods, assumptions you make, and describe the possible advantages/disadvantages, problems.
- Carefully reflect on the parameterization of your models, as well as on the nature of the covariates: which ones will be used as time-varying, which will be used as baseline-characteristics? Is a data transformation necessary? Will you use the time variable as categorical i.e., visits or as continuous i.e., years/months since first visit?
- Discuss any issues related to missing values. Under which assumptions about the missing data are your results valid? Are there alternative approaches to consider other than the ones mentioned above? Would you recommend multiple imputation?
- Do you have any recommendations with respect to future similar experiments?
- Include as an appendix the R-code that reproduces your analyses.
- You are encouraged to use R markdown to prepare your report. In this case you may also submit the .Rmd file that reproduces your report and analyses.

Good luck!